

CSM V3 — CIRCULAR SOCK MACHINE

Complete Machine Architecture — Engineering Overview

14 Modules • Layer 1 Precision • Layer 2 Structural • Layer 3 Automation

MASTER DATUM
Z = 230 mm

(Top of aluminum plate,
ICD R7 invariant B3)

LAYER KEY

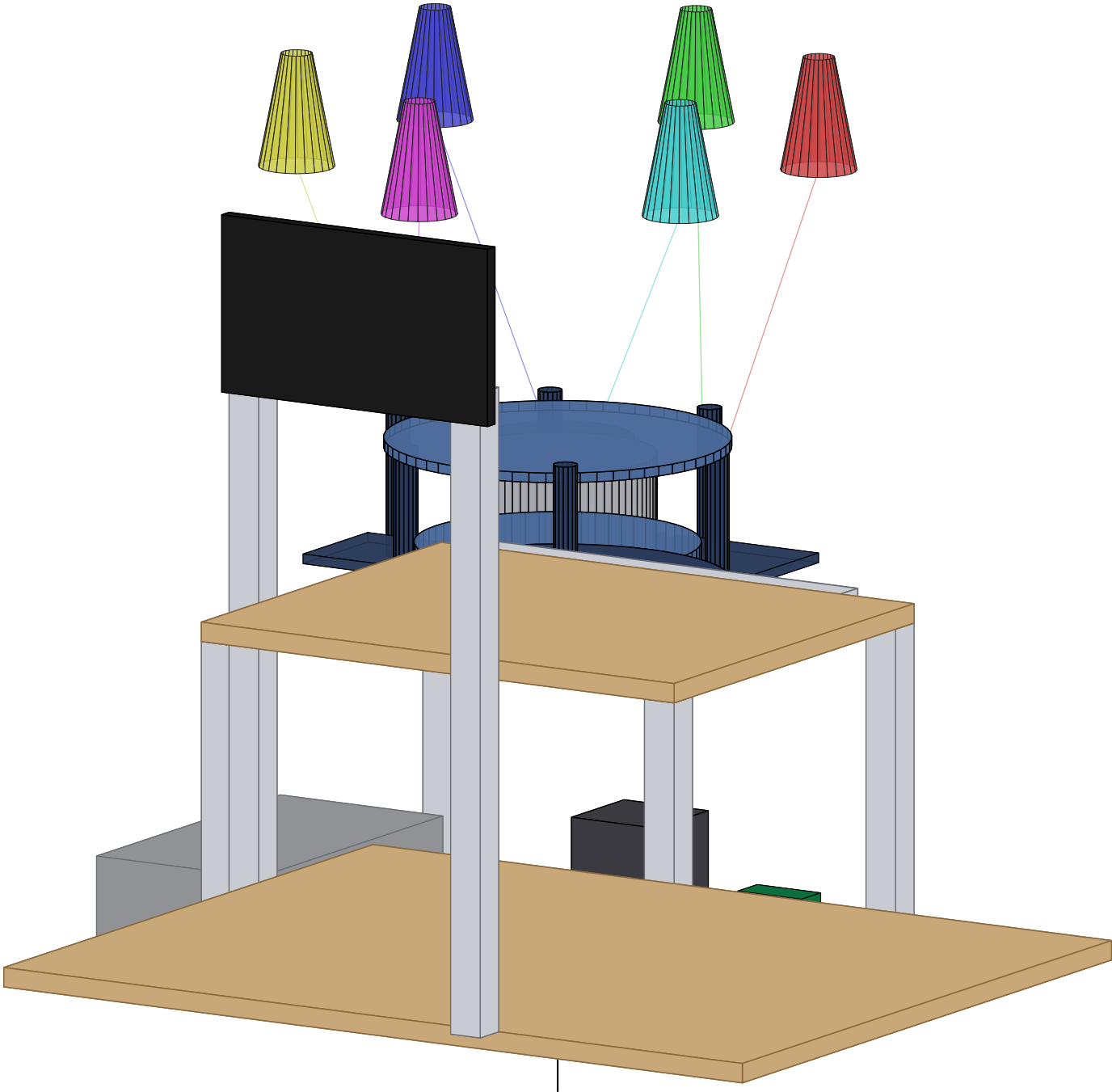
Wood (Baltic Birch 12 mm)

6061-T6 Aluminum (plate + 2020)

PETG printed (Phase 1)

Steel / motor

Electronics PCB



Module 1	72-slot Knitting Cylinder V3.1
Module 2	Cam Ring V6.5
Module 3	Sinker Ring V1.2.1
Module 4	Cassette Base + Retainer + 6× Spacers
Module 5	6× Feeder Assembly (servo-driven, ceramic pigtails)
Module 6	Drive — NEMA 23 + HG5 5:1 gearbox + HTD belt
Module 7	Bearing & Shaft Stack — floating-top / fixed-bottom
Module 8	Take-Down — Phase 1 hook / Phase 2 dual roller
Module 9	Frame — wood base 500×400×12 + 4× 2020 + Ø250 plate
Module 10	HMI — dual 2020 mast + 7" touchscreen + Pi 4
Module 11	Electronics — S-250-24 PSU + Mega + 2× LM2596 + TB6600
Module 12	Operator Panel — E-stop + AC inlet + fuse
Module 13	Ribber Disk — provisioned for Phase 2
Module 14	Yarn Path — 6 cones + ceramic pigtails